WORKED SOLUTION

STEP I 2007, Question 13

Probability and Statistics • Conditional Probability

STEP I 2007, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/07-s1-q13

This question has a full guided treatment in *Beyond STEP*, Section 12.1, *Precise Conditioning*: https://beyondstep.keithmcnulty.org/ch12/conditioning.html

The key idea

Although the discs come out one at a time, every set of four discs is equally likely, so most of the work is counting combinations. The interest is in the conditions. Knowing *which* disc came first, or simply *that* a particular disc was taken, leaves the other three discs as a random choice from the ten that remain. Knowing only that *some* numbered disc was taken is a different kind of information altogether, and the last two parts are there to show the difference.

Solution

Let N be the number of numbered discs among the four taken. Since the bag is shaken, every ordered sequence of four distinct discs is equally likely, and so every unordered set of four discs is equally likely too. There are

$$\binom{11}{4} = 330$$

such sets. A set with exactly k numbered discs is made by choosing k of the 5 numbered discs and $4-k$ of the 6 blanks, so

$$\mathrm{P}(N=k) = \frac{\binom{5}{k}\binom{6}{4-k}}{330}.$$

Part (i)

All four numbered means $k=4$:

$$\mathrm{P}(N=4) = \frac{\binom{5}{4}\binom{6}{0}}{330} = \frac{5}{330} = \frac{1}{66}.$$

Equivalently, drawing in order, $\frac{5}{11} \cdot \frac{4}{10} \cdot \frac{3}{9} \cdot \frac{2}{8} = \frac{1}{66}$.

Part (ii)

Let T_1 be the event that disc "3" is taken first. Given T_1, the shaken bag now holds 10 discs, 4 numbered and 6 blank, and the next three discs are a random selection from these. All four are numbered exactly when these three are all numbered, so

$$\mathrm{P}(N = 4 \mid T_1) = \frac{\binom{4}{3}}{\binom{10}{3}} = \frac{4}{120} = \frac{1}{30}.$$

Part (iii)

Still given T_1, we need exactly one more numbered disc among the three drawn from the remaining ten:

$$\mathrm{P}(N = 2 \mid T_1) = \frac{\binom{4}{1}\binom{6}{2}}{\binom{10}{3}} = \frac{4 \times 15}{120} = \frac{1}{2}.$$

Part (iv)

Now let T be the event that disc "3" is taken at some point.

? **Ask yourself** Does it matter whether disc "3" came first or at some other draw?

No. Given T, the four discs taken are disc "3" together with three others, and by symmetry every set of three from the other ten discs is equally likely. That is exactly the situation of part (iii), where the order of the draws never mattered. So the answer is again $\frac{1}{2}$.

To confirm this directly: $\mathrm{P}(T) = \binom{10}{3}/\binom{11}{4} = \frac{120}{330} = \frac{4}{11}$, and the sets containing disc "3", one other numbered disc and two blanks number $\binom{4}{1}\binom{6}{2} = 60$, so

$$\mathrm{P}(N = 2 \mid T) = \frac{60/330}{120/330} = \frac{1}{2}.$$

Part (v)

Let F be the event that the first disc is numbered. Split F according to which number appears on the first disc: F is the disjoint union of the five events "disc j is taken first", $j = 1, \ldots, 5$, each of probability $\frac{1}{11}$. The argument of part (iii) did not use the fact that the number was 3, so for each j

$$\mathrm{P}(N = 2 \mid \text{disc } j \text{ first}) = \tfrac{1}{2}.$$

Hence, by the law of total probability,

$$\mathrm{P}(N = 2 \mid F) = \sum_{j=1}^{5} \frac{1/11}{5/11} \cdot \frac{1}{2} = \frac{1}{2}.$$

(Directly: given F, the other three discs are a random three from 4 numbered and 6 blank, which is part (iii) again.)

Part (vi)

Let A be the event that at least one numbered disc is taken, so A is the event $N \geq 1$. This is *not* the same sort of information as before: it does not single out a disc and leave the rest random. Instead we use the definition of conditional probability. Since $N = 2$ implies A,

$$\mathrm{P}(N = 2 \mid A) = \frac{\mathrm{P}(N = 2)}{\mathrm{P}(N \geq 1)}.$$

Ask yourself Why can't we just say "one numbered disc is known, so the other three are random from the remaining ten"?

Because we don't know *which* disc is the known one. A set with two numbered discs satisfies A in the same way as a set with four; there is no single disc we can set aside. So the condition reweights the sets according to N itself, and we must count.

We have

$$\mathrm{P}(N = 2) = \frac{\binom{5}{2}\binom{6}{2}}{330} = \frac{10 \times 15}{330} = \frac{150}{330}, \qquad \mathrm{P}(N = 0) = \frac{\binom{6}{4}}{330} = \frac{15}{330},$$

so $\mathrm{P}(N \geq 1) = \frac{315}{330}$ and

$$\mathrm{P}(N = 2 \mid A) = \frac{150}{315} = \frac{10}{21}.$$

Checks

The distribution of N should add up: the counts $\binom{5}{k}\binom{6}{4-k}$ for $k = 0, \ldots, 4$ are $15, 100, 150, 60, 5$, which total 330. In parts (ii) and (iii) the conditional distribution of the number of further numbered discs has counts $20, 60, 36, 4$ out of 120, again summing correctly. We also checked all six answers ($\frac{1}{66}$, $\frac{1}{30}$, $\frac{1}{2}$, $\frac{1}{2}$, $\frac{1}{2}$, $\frac{10}{21}$) by listing all $11 \times 10 \times 9 \times 8 = 7920$ ordered draws with exact fractions. Note that $\frac{10}{21} \approx 0.476$ is a little below $\frac{1}{2}$, as it should be: conditioning on $N \geq 1$ only removes the $N = 0$ case, so it leaves more weight on $N = 1$ than knowing a particular disc does.

Common slips

- Treating parts (iv) and (vi) as the same kind of condition. "Disc 3 is taken" names a disc; "a numbered disc is taken" does not, and the answers differ.
- In part (vi), forgetting that the condition is $N \geq 1$ and dividing by $\mathrm{P}(N = 1)$ or by $\frac{5}{11}$ instead.
- Counting ordered draws in the numerator and unordered sets in the denominator. Choose one and stick to it.

Beyond STEP

The contrast between parts (v) and (vi) is the same trap as the famous "boy or girl" paradox: being told that a *specific* child is a boy gives a different answer from being told that *at least one* is. At university this is made precise by treating conditioning on an event as restricting and renormalising the probability measure, and the counts here are the hypergeometric distribution. Grimmett and Welsh, *Probability: An Introduction*, and Blitzstein and Hwang, *Introduction to Probability*, both discuss these paradoxes carefully.

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